# A genuine descent direction on the harmonic-alignment branch

Trace-free Ricci criticality and almost-Kähler equality geometry

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**Abstract**

We study the unrestricted functional $P(g) = \int |\operatorname{Ric}_g^\circ|^2$ on closed four-manifolds. For a constant-scalar almost-Kähler metric with $J$-invariant Ricci tensor, we derive the exact first variation $DP_g[\operatorname{Ric}_g^\circ] = -sP(g)/2$. Thus every non-Einstein member with $s \neq 0$ has an explicit volume-preserving descending path, without a Kähler, symmetry, period, or spectral assumption. Applied to the harmonic-flux equality theorem in the supplied October 8 manuscript, this excludes its entire non-Einstein equality branch from being a smooth global $P$-minimizer. A related identity retains the amplitude correlation in the more general harmonic-product case. These results provide an additional obstruction to proposed minimizers; they do not establish smooth attainment on the Godeaux blowup or solve the general non-Bach-flat minimizer problem.

## 1 Functional and conventions

Let $(M, g)$ be a smooth closed connected oriented four-manifold and put

$$Z = \operatorname{Ric} - \frac{s}{4}g, \qquad P(g) = \int_M |Z|^2 \, dV_g.$$

Two-forms have the form norm, so $|e^{12} + e^{34}|^2 = 2$; they are identified with skew endomorphisms when composed. Weyl curvature is an endomorphism of $\Lambda^2$. Write

$$\mathscr{R}(H, H) = R_{ikjl}H^{ij}H^{kl}, \qquad R_{ikjl}g^{kl} = \operatorname{Ric}_{ij},$$

with positive sectional curvature convention. The rough Laplacian $\nabla^*\nabla$ and scalar Laplacian $\Delta_0 = d^*d$ are nonnegative. A constant-scalar metric has $\operatorname{div} Z = 0$ by contracted Bianchi.

The classical gradients of $\int |\operatorname{Ric}|^2$ and $\int s^2$ in [1, (2.5), (2.9)] give, at constant scalar curvature,

$$\mathcal{G}_P = \nabla^*\nabla Z - 2\mathscr{R}(Z) + \tfrac{1}{2}|Z|^2 g. \tag{1.1}$$

Here $DP_g[H] = \int \langle \mathcal{G}_P, H\rangle \, dV_g$. Consequently

$$DP_g[Z] = \int_M \left(|\nabla Z|^2 - 2\mathscr{R}(Z, Z)\right) dV_g. \tag{1.2}$$

The direction $Z$ is fixed at the base metric for this derivative. No differentiation of a choice of a field along a metric family is used. The conformal trace equation for a closed $P$-critical metric is $\Delta_0 s = 0$, so every such metric has constant scalar curvature. Scale invariance makes criticality on unit-volume metrics equivalent to unrestricted criticality.

## 2 The opposite-duality product identity

**Lemma 2.1.** *Let $\phi \in \Lambda^+$, $\psi \in \Lambda^-$, and $H = \phi \circ \psi$. Then $H$ is symmetric and trace-free and*

$$|H|^2 = |\phi|^2|\psi|^2, \tag{2.1}$$

$$\mathscr{R}(H, H) = |\psi|^2\langle W^+\phi, \phi\rangle + |\phi|^2\langle W^-\psi, \psi\rangle - \frac{s}{12}|\phi|^2|\psi|^2. \tag{2.2}$$

*For smooth forms put $f = |\phi|^2$ and $h = |\psi|^2$. Then*

$$|\nabla H|^2 = h|\nabla\phi|^2 + f|\nabla\psi|^2 + \tfrac{1}{2}\langle df, dh\rangle. \tag{2.3}$$

*Proof.* Opposite-duality skew endomorphisms commute. Their composition realizes an isometry $\Lambda^+ \otimes \Lambda^- \simeq S_0^2$ with these norms. This proves (2.1) and, after differentiating the product, (2.3). Explicitly the cross term is $2\sum_i \langle \nabla_i\phi, \phi\rangle\langle\psi, \nabla_i\psi\rangle$.

At a point with both forms nonzero, choose an oriented orthonormal frame such that $\phi = a(e^{12} + e^{34})$ and $\psi = b(e^{12} - e^{34})$. Then $H = ab\,\mathrm{diag}(-1, -1, 1, 1)$ and

$$\mathscr{R}(H, H) = 2a^2b^2(K_{12} + K_{34} - K_{13} - K_{14} - K_{23} - K_{24}).$$

Using $s = 2\sum_{i<j} K_{ij}$ and the two diagonal curvature blocks $W^{\pm} + sI/12$ gives (2.2). The trace-free Ricci block drops out of this contraction; it has not been assumed zero. The identities extend over zero sets by continuity. □

**Proposition 2.2** (Harmonic-product first variation)**.** *Suppose $s$ is constant and there are smooth harmonic forms $\phi \in \Omega^+(M)$ and $\psi \in \Omega^-(M)$ such that $Z = \phi \circ \psi$. Then*

$$\boxed{DP_g[Z] = -\frac{s}{2}P(g) - \frac{1}{2}\int_M \langle d|\phi|^2, d|\psi|^2\rangle\, dV_g.} \tag{2.4}$$

*In particular, if either harmonic form has constant length, then $DP_g[Z] = -sP(g)/2$.*

*Proof.* The harmonic two-form Weitzenböck equations are

$$\nabla^*\nabla\phi = 2W^+\phi - \frac{s}{3}\phi, \qquad \nabla^*\nabla\psi = 2W^-\psi - \frac{s}{3}\psi.$$

Thus

$$|\nabla\phi|^2 = 2\langle W^+\phi, \phi\rangle - \frac{s}{3}f - \tfrac{1}{2}\Delta_0 f, \quad |\nabla\psi|^2 = 2\langle W^-\psi, \psi\rangle - \frac{s}{3}h - \tfrac{1}{2}\Delta_0 h.$$

Substitute these in (2.3), subtract twice (2.2), and integrate. The scalar term is $-sfh/2$. The amplitude terms are

$$-\tfrac{1}{2}\int (h\Delta_0 f + f\Delta_0 h) + \tfrac{1}{2}\int \langle df, dh\rangle = -\tfrac{1}{2}\int \langle df, dh\rangle.$$

Equation (1.2) now proves the assertion. All fields are smooth on the original closed manifold, so there are no puncture or end terms in this integration. □

For a nonzero negative-scalar $P$-critical metric in this general harmonic-product class, (2.4) requires

$$\int_M \langle d|\phi|^2, d|\psi|^2\rangle\, dV_g = |s|P(g) > 0. \tag{2.5}$$

This is a necessary condition, not an exclusion of every Einstein–Maxwell geometry. When neither amplitude is constant, its correlation cannot be omitted.

## 3 An unrestricted descent for the almost-Kähler branch

**Theorem 3.1.** *Let $(M^4, g, J, \Omega)$ be closed almost-Kähler, with constant scalar curvature $s$ and $J$-invariant Ricci tensor. Then*

$$\boxed{DP_g[Z] = -\frac{s}{2}P(g).} \tag{3.1}$$

*If $s \neq 0$ and $g$ is not Einstein, it is not even a stationary point of the unrestricted $P$ functional. An explicit descending family is*

$$g_t(X, Y) = g\big(\exp(t\,\mathrm{sgn}(s)Z^\sharp)X, Y\big), \qquad t > 0 \text{ small.} \tag{3.2}$$

*It preserves the volume form pointwise and satisfies*

$$P(g_t) = P(g) - \frac{|s|}{2} P(g)t + O_g(t^2). \tag{3.3}$$

*Moreover*

$$\|\mathcal{G}_P\|_{L^2} \geq \frac{|s|}{2}\sqrt{P(g)}. \tag{3.4}$$

*Proof.* Use the convention $\Omega^\sharp = J$; changing its sign has no effect on the argument. The symmetric endomorphism $Z^\sharp$ commutes with $J$ and is trace-free. Therefore $\psi = -JZ^\sharp$ is an anti-self-dual two-form and $Z = \Omega \circ \psi$.

The almost-Kähler form is self-dual, closed, and of constant norm $\sqrt{2}$, hence harmonic. For forms of opposite duality the product rule, when the first form is closed, gives

$$\operatorname{div}(\Omega \circ \psi) = \Omega(d^*\psi).$$

Indeed $d^*\Omega = 0$; the remaining differentiated anti-self-dual endomorphisms commute with $\Omega$, yielding the displayed identity. Contracted Bianchi gives $\operatorname{div} Z = 0$. Since $J$ is invertible, $d^*\psi = 0$; anti-self-duality gives $d\psi = 0$. Thus $\psi$ is harmonic globally, including at its zeros. This also recovers the usual harmonic primitive-Ricci-form fact without assuming that $J$ is integrable.

Apply Proposition 2.2 with $\phi = \Omega$, whose norm is constant. The derivative of (3.2) at zero is $\operatorname{sgn}(s)Z$. The endomorphism exponential is positive, and its determinant is one because $\operatorname{tr} Z = 0$. This proves (3.3) and volume preservation. The family need not remain in the almost-Kähler subspace: Q3 asks about all metrics. Finally, Cauchy–Schwarz applied to $|DP_g[Z]| = (|s|/2)P(g)$ proves (3.4). □

**Corollary 3.2.** *A smooth closed $P$-critical metric which admits an almost-Kähler structure with invariant Ricci tensor is Einstein if $s \neq 0$. If $s = 0$, it is Bach-flat. In particular no metric in this class answers the non-Bach-flat global-minimizer problem.*

*Proof.* Criticality first makes $s$ constant. For $s \neq 0$, (3.1) forces $P = 0$. For $s = 0$, write

$$P = \left(\int |\operatorname{Ric}|^2 - \frac{1}{3}\int s^2\right) + \frac{1}{12}\int s^2.$$

The last functional has zero gradient at $s = 0$. The first is, up to a topological term and a nonzero factor, the Weyl functional. Thus criticality is Bach-flatness [1, Section 1.4]. □

## 4 Application to the supplied harmonic-flux equality theorem

For a nonzero harmonic anti-self-dual form $h$, set

$$H = \|h\|_2^2, \quad \mathfrak{F}_h = \sqrt{2}\int_M |B^*h|\, dV_g, \quad L_h = \frac{P/2 - \mathfrak{F}_h^2/H}{4\pi^2}, \qquad |Z|^2 = 4|B|^2.$$

The supplied manuscript [2, Section 3.2] proves that at constant scalar curvature, $L_h = 0$ implies either Einstein geometry, or a global almost-Kähler structure with invariant Ricci tensor and primitive Ricci form a constant multiple of $h$. Its bounded harmonic form extends across the zeros of $h$ by the stated removability argument.

Combining that result with Theorem 3.1, rather than solving the negative Blair–Ianus integrability problem, gives

$$\boxed{s \neq 0,\ P(g) > 0,\ L_h(g) = 0 \quad \Longrightarrow \quad g \text{ is not an unrestricted local } P\text{-minimizer.}} \tag{4.1}$$

In the normalization $s = -4$, the actual path $g_t = g(e^{-tZ^\sharp}\cdot, \cdot)$ has $DP_g[-Z] = -2P(g) < 0$. Replacing $g_t$ by its negative Yamabe representative, and rescaling if desired, cannot increase $P$; hence the decrease is not an artifact of leaving the constant-scalar locus.

If a non-Einstein, negative-scalar smooth global minimizer $g_*$ exists, then for every nonzero harmonic ASD form $h$ one must have $L_h(g_*) > 0$. In fact, provided $b^- > 0$,

$$\min_{\substack{h \in \mathcal{H}^-(g_*) \\ \|h\|_2 = 1}} L_h(g_*) > 0. \tag{4.2}$$

The sphere is compact in a fixed finite-dimensional vector space, and $h \mapsto \int |B^* h|$ is continuous. A zero minimum would contradict (4.1). This positive number is allowed to depend on $g_*$; it is not uniform over degenerating geometries.

The latest product calculation in [2, Section 4] concerns $P/(8\pi^2) - |a_g^-|^2$ with a moving harmonic period. The present calculation concerns $P$ itself under unrestricted metric variations. These stationarity conditions are different. In particular the new conclusion does not contradict stationarity of the corrected functional at a non-Einstein cscK metric.

## 5 The precise Q3 conclusion

Keep the Godeaux blowup $M = X \# \overline{\mathbb{CP}}^2$ used in the previous continuation, with the inherited estimates

$$\frac{16\pi^2}{3} \leq P_*(M) < 8\pi^2,$$

and negative Yamabe constant for every conformal class. If $P_*(M)$ is attained smoothly, the attaining metric is non-Einstein and non-Bach-flat. This reduction, the strict upper construction, and the previous negative-lattice exclusion are project inputs [3].

The present result adds a definitive restriction on such an attaining metric: it cannot belong to the entire harmonic-flux equality branch, even if its almost-complex structure is nonintegrable. There is no need to decide integrability of that branch before excluding it from Q3. The proof supplies a signed, admissible metric variation, not only a nonnegative decomposition.

It does not exclude arbitrary ALE or partially collapsed minimizing limits. It also does not imply that a minimizing sequence has $L_h \to 0$. Smooth attainment on the fixed manifold, or another complete example for Q3, remains to be proved. The four public questions are not claimed solved by this note.

## References


[1] M. T. Gursky and J. A. Viaclovsky, *Rigidity and stability of Einstein metrics for quadratic curvature functionals*, arXiv:1105.4648v2 (2013). `https://arxiv.org/abs/1105.4648`. The variational conventions used here are equations (2.5) and (2.9).

[2] Qiushi, *Einstein Gaps on Four-Manifolds: Subsequent Mathematical Developments*, 8 October 2026, `03_Subsequent_Mathematical_Developments(1).tex`. Harmonic-flux decomposition, equality theorem, and constrained product variation are the supplied inputs discussed in Section 4.

[3] *Global period constraints for the Q3 minimizing problem*, 8 October 2026, `Q3_Global_Period_Constraints.tex`; and the earlier supplied four-problem synthesis. These supply the fixed attainment problem, not an existence theorem.